

Research Project : Compound Climate Events and Health

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My academic and professional background



Baccalaureate C

2016



Bachelor's degree in Physics

2016-2020



Professional Teaching
certificate (Grade II) in Physics

2020-2022



Leadership training

2022-2023



Master's degree in
Mathematical Sciences

2023-2024

Helping out-of-school or unemployed
youth through various free training programs

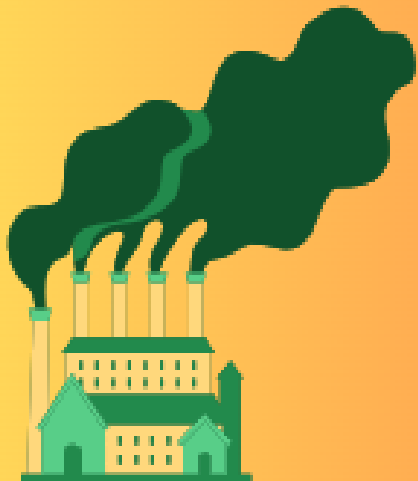
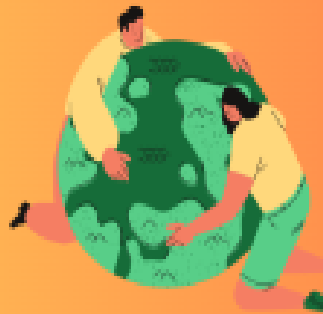


Physics, Mathematics, and Entrepreneurship
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Published research article

Impact of CLIMATE CHANGE



Extremes events and socio-economic impacts in central Africa: a CMIP6-based analysis of projections

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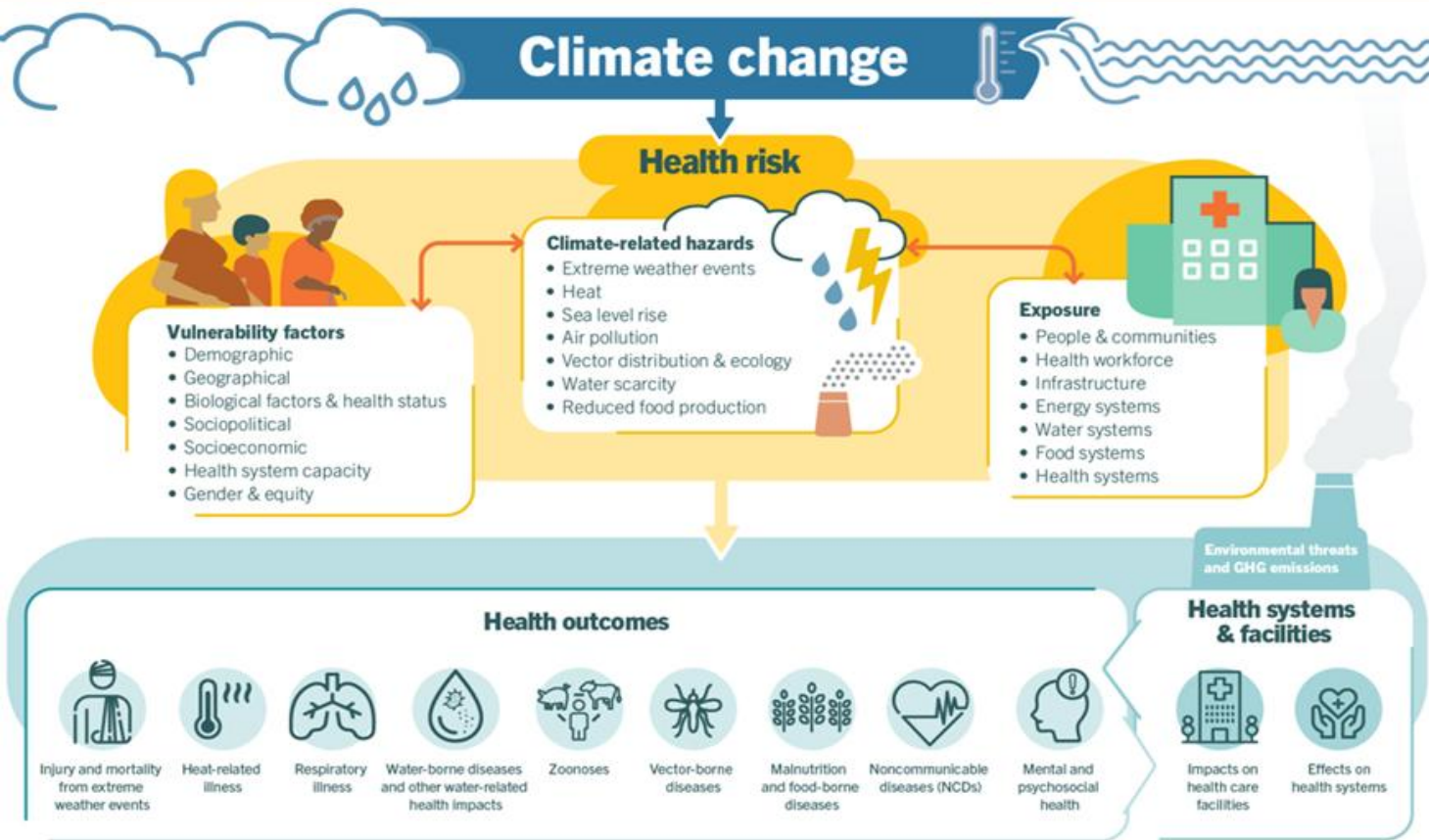
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Abstract

Extreme events such as floods, droughts, and heatwaves profoundly impact all socio-economic sectors in Central Africa. Previous studies often focus on how extreme events respond to global warming without contextualizing these results within socio-economic sectors. In contrast, the present study uses data from 13 global climate models participating in the sixth phase of the Coupled Model Intercomparison Project (CMIP6) to explore the impacts of global warming-induced changes in extreme events across various socio-economic sectors. Six indices defined by the Expert Team on Climate Change Detection and Indices (ETCCDI) are employed for the purpose. Analyses are conducted over two time frames, namely 2030–2059 for the near future and 2071–2100 for the far future in comparison to the historical period 1985–2014, under the low SSP126 and high SSP585 emission scenarios. The results indicate enhancements in dry spells and weakening in wet spells in response to global warming. Simultaneously, an increase in annual total precipitation is expected, in association with intensification in heavy precipitation days and daily precipitation intensity. Temperature-based indices exhibit a decreasing trend in the total number of cold days per year and an increasing trend in the number of hot days, with more intense changes under the unmitigated SSP585 scenario. The low-emission, highly mitigated SSP126 scenario demonstrates its effectiveness in limiting the worsening of projected conditions compared to SSP585. Discussing the potential socio-economic risks associated with these changes highlights the urgent need to formulate robust policies to mitigate underlying hazards, as they could lead to challenges such as food insecurity, heat and humidity-related illnesses, population impoverishment, market inflation, and social instability.

Keywords Extreme events · CMIP6 · Precipitation · Temperature · Future climate projections · Central Africa



Context

The Sixth Assessment Report of the IPCC warns that compound climate events (CCEs) represent a major emerging threat to human societies, with amplified, cumulative, and often unpredictable risks, especially in regions already characterized by high vulnerability. This situation is particularly concerning for sub-Saharan Africa, which has been identified as one of the world's main climate vulnerability hotspots. In this region, repeated exposure to extreme events is combined with fragile health systems, limited infrastructure, and socio-economic constraints (Thornton et al., 2014; Engelbrecht et al., 2015), greatly increasing health risks, especially for children (Dimitrova et al., 2022).

Schéma conceptuel de l'OMS illustrant comment le changement climatique constitue un risque pour la santé via des aléas climatiques (événements extrêmes, chaleur, élévation du niveau de la mer, pollution de l'air, perturbation écologique, etc.) et des facteurs d'exposition (populations, infrastructures, systèmes vitaux), modulés par des facteurs de vulnérabilité (démographie, statut socio-économique, état de santé, capacités du système de santé, etc.). Ces interactions conduisent à de multiples issues sanitaires : traumatismes et mortalité liés aux événements extrêmes, maladies liées à la chaleur, pathologies respiratoires, maladies d'origine hydrique ou alimentaire, zoonoses, malnutrition, maladies non transmissibles, troubles mentaux, ainsi qu'à des impacts sur les établissements et services de santé (dégâts aux infrastructures, surcharge des systèmes de soins)



Earthquakes



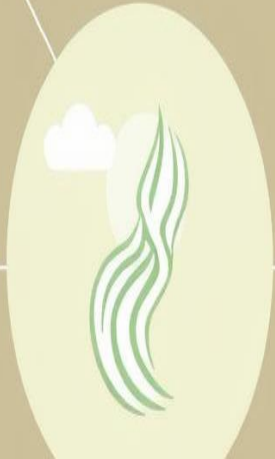
Floods



Storms



Hurricanes



Droughts



Context

Extreme climate events do not always occur in isolation. The climate system often presents complex configurations where multiple hazards interact across time or space. These are known as Compound Climate Events (CCEs). In the context of climate change, such events are expected to become more frequent, more intense, and more interconnected.

Definition of Compound Events

Combination of multiple climate hazards and/or drivers jointly contributing to environmental or societal risk

Classified into four types ((Zscheischler et al. (2020))

Preconditioned / Triggered

A prolonged hazard creates antecedent conditions that exacerbate the impacts of subsequent events (e.g., heavy rainfall following a long drought triggering a landslide)

Multivariate

Co-occurrence of multiple climatic or environmental drivers in the same geographical region, leading to a combined impact

Temporally Compounded

A sequence of hazards affects a given region over a short time period, amplifying their cumulative effects

Spatially Compounded

Several connected regions are affected either by the same hazard or by different hazards within a limited time frame

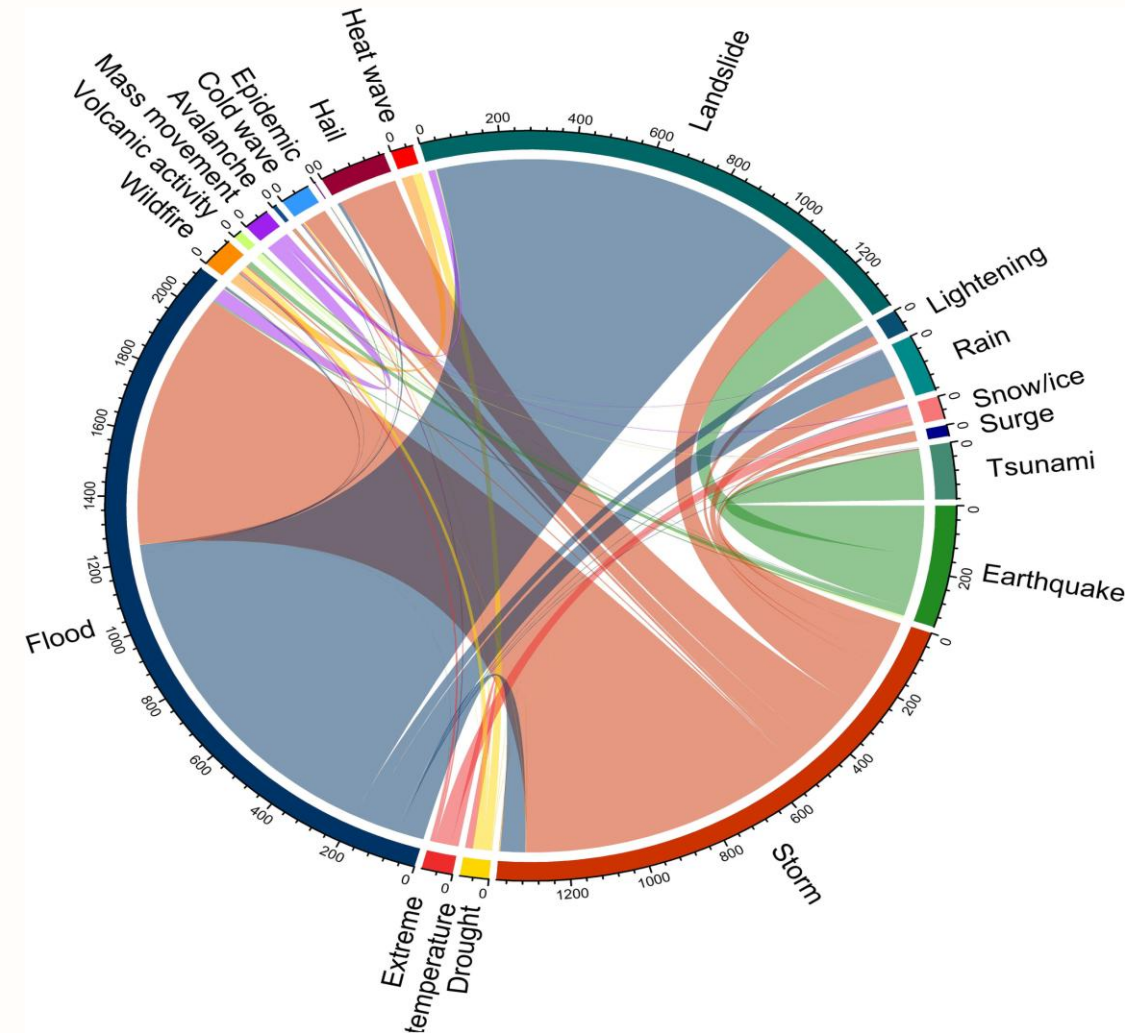
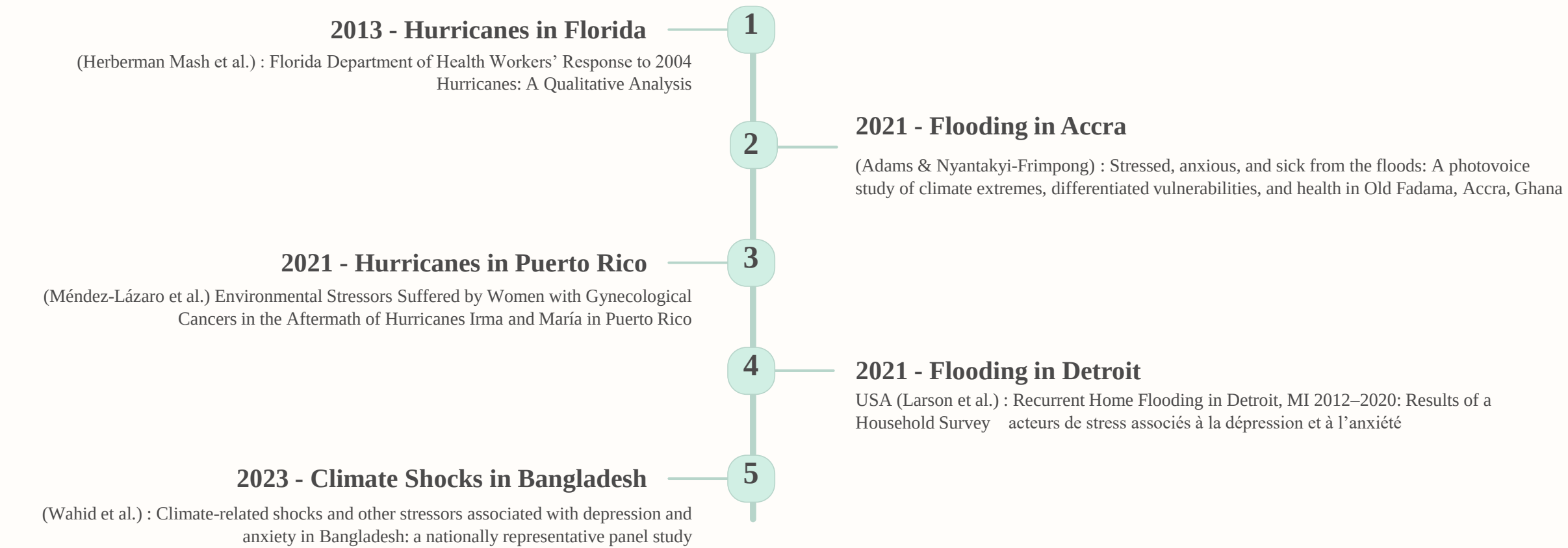


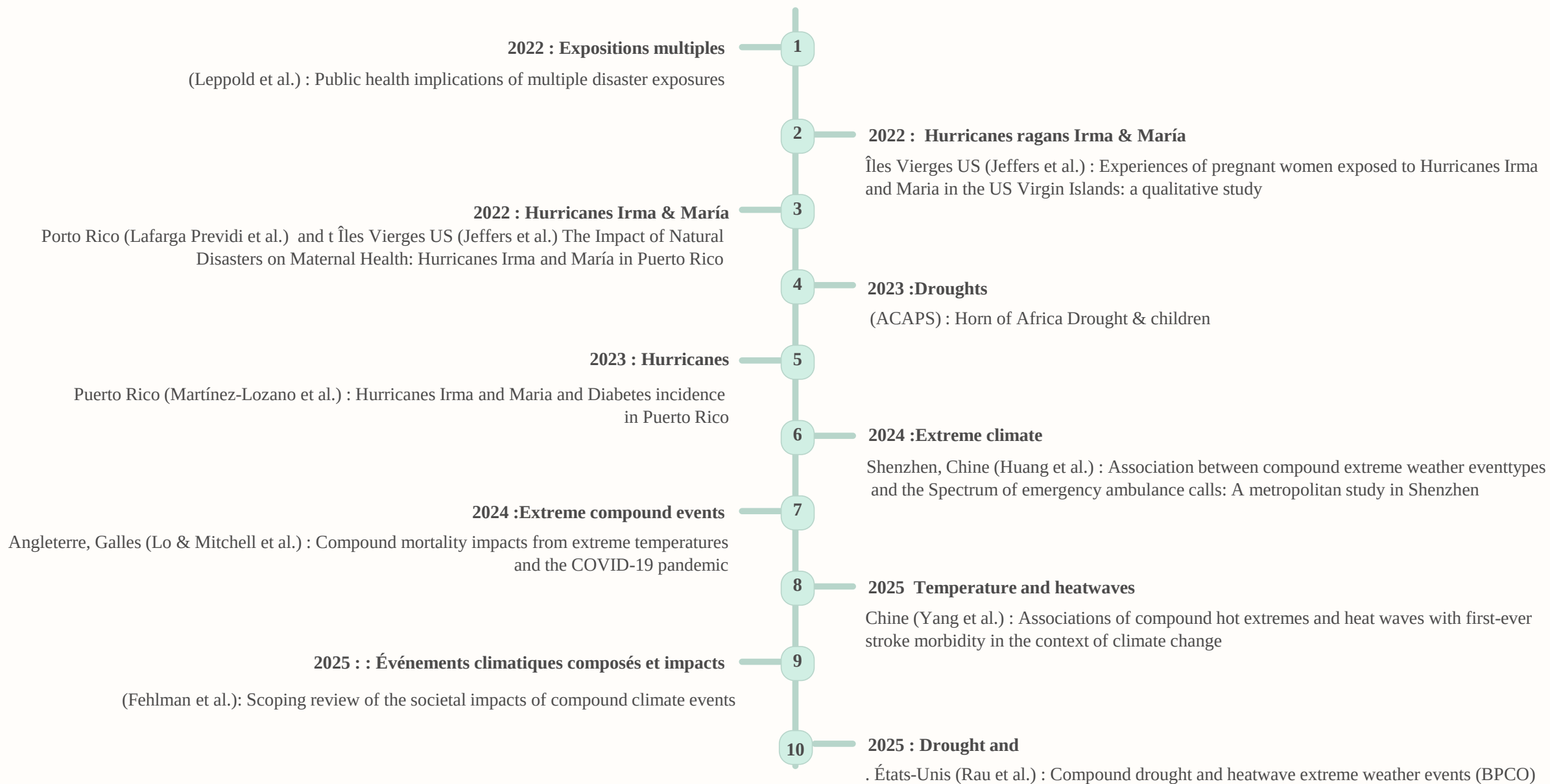
Illustration de la relation proportionnelle et directionnelle entre différents aléas (Lee et al., 2024)

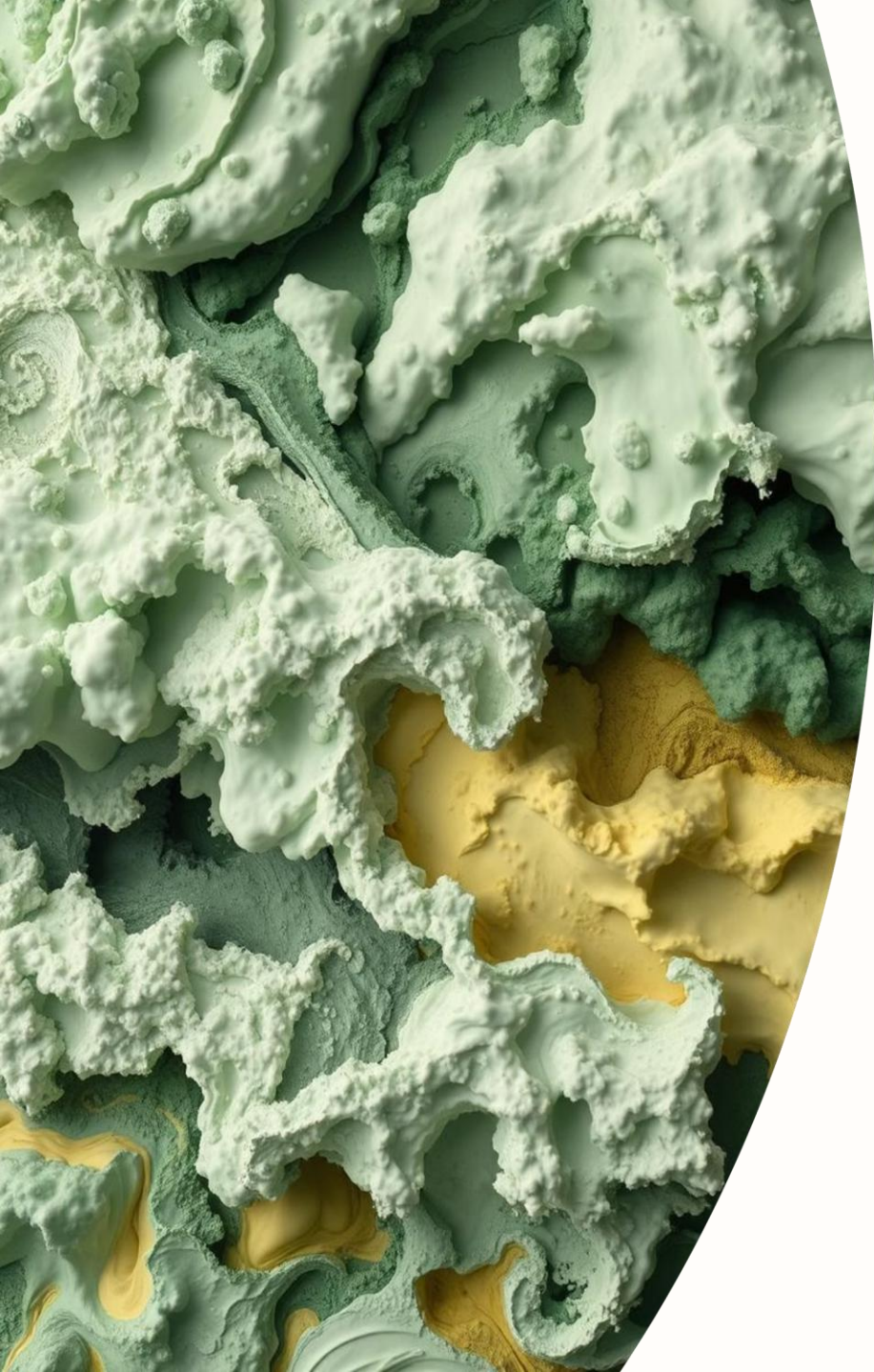
Literature Review on CCEs and Health

Previous studies, like Dimitrova et al. (2022), have already shown links between isolated hazards such as heatwaves and health outcomes, particularly child health in vulnerable communities in West Africa. However, most of the existing literature still focuses on single hazards, despite the growing interest in Compound Climate Events (CCEs) in recent years.

A recent scoping review (Fehlman et al.,2025) identified only 28 studies (up to early 2023) that explicitly focused on the societal impacts of compound climate events. Among these, only five specifically addressed public health impacts.







Problem Statement

In the face of accelerating climate change, it is crucial to understand how compound climate hazards crossing critical thresholds in space and time influence the dynamics of health outcomes in African countries.

Research Objectives

- Describe the incidence and distribution of extreme hazards
- Analyze the spatial combined effect of hazards
- Study the sequential or temporal effect of hazards



Potential Data Sources



Climate Data

- CHIRPS: High-resolution precipitation estimates
- ERA5: Reanalysis dataset (temperature, humidity, wind, etc.)



Health Data

- DHS (Demographic and Health Surveys): Geolocated health and demographic indicators (nutrition, child health, household assets, etc.)



Disaster & Impact Databases

- EM-DAT (CRED): Global disaster database with multi-hazard events



Conclusion

This research project is still in its exploratory phase. We are currently reviewing the literature and exploring available data sources.